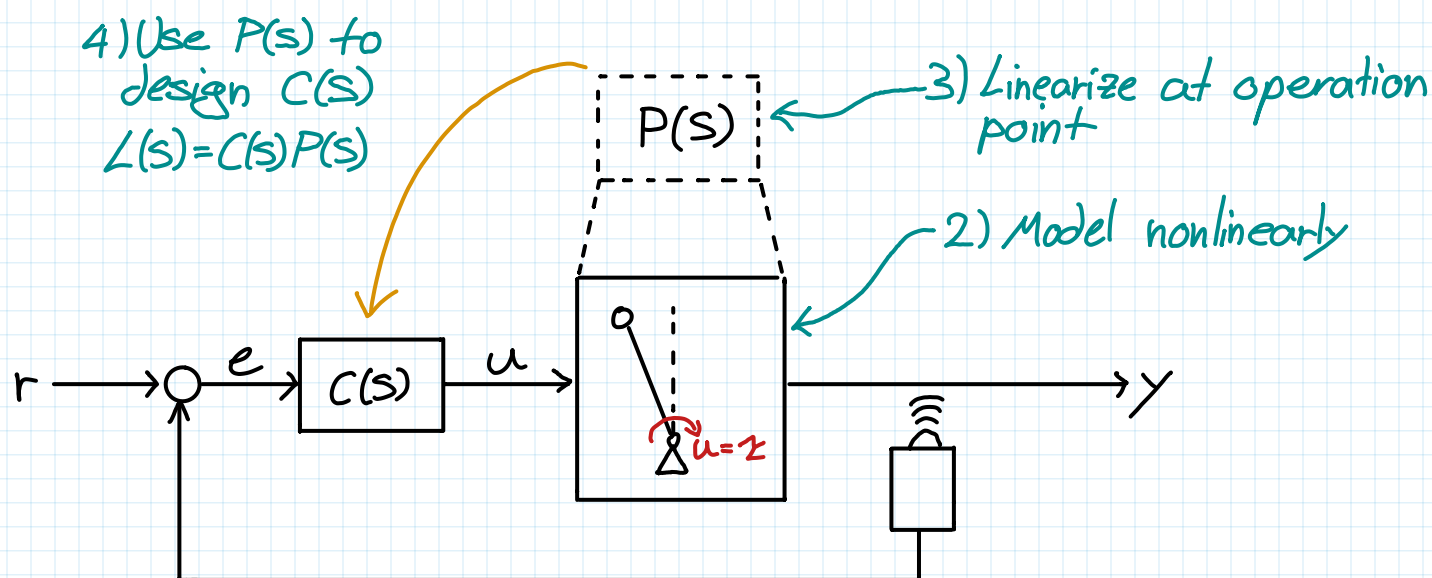


Previously in CSI

Task: Design controller for a system such that the closed-loop:

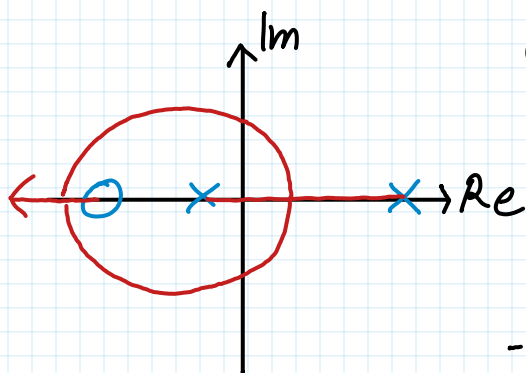
- Is stabilized (I)
- Is robust (II)
- Rejects noise and disturbances (III)

1) Define architecture:

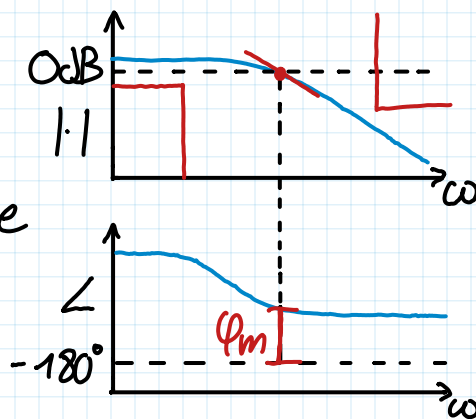


5) Use knowledge of closed-loop structure $T(s)$ to translate closed-loop requirements (I - III) into open-loop ones.

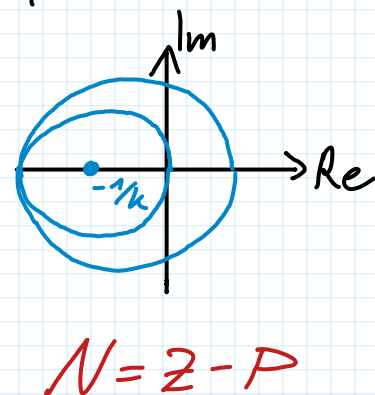
• Root Locus



• Bode Plot



• Nyquist Plot

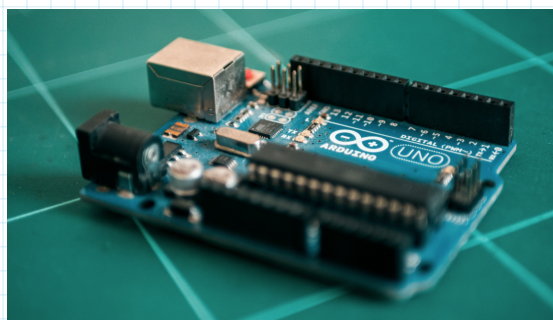


6) Build a controller using:

$$\cdot k \quad \cdot 1/s \quad \cdot \frac{s/a+1}{s/b+1}, 0 < a < b \quad \cdot \frac{s/a+1}{s/b+1}, 0 < b < a$$

(Or just use PID $C_{PID}(s) = k_P + k_D s + k_I \frac{1}{s}$ and tune gains until satisfactory response)

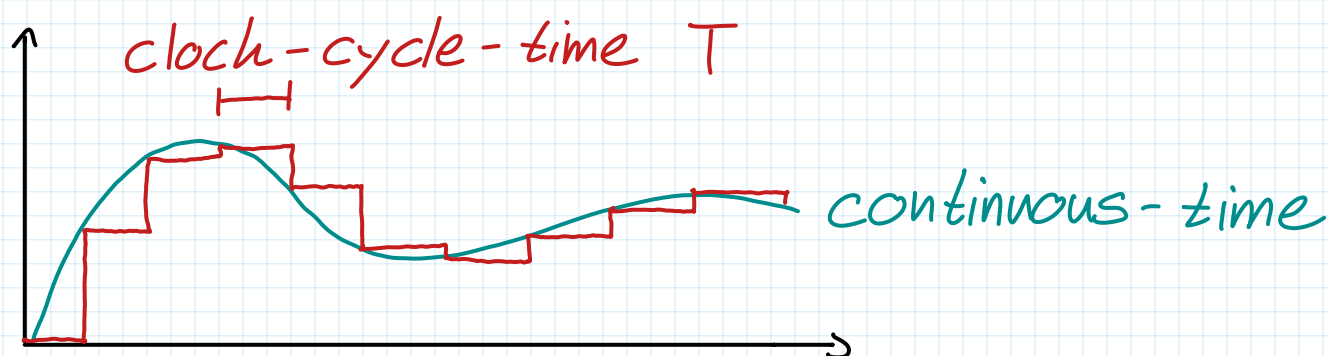
7) Implement!! But how?



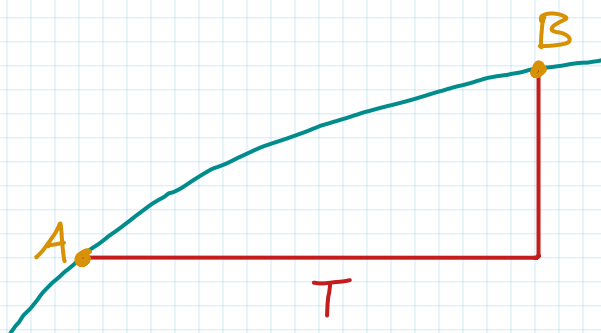
Write C-code that implements the controller on a microcontroller.

Problem: These operate in discrete time!

=> How do we make this work if all our design efforts have been in continuous time?

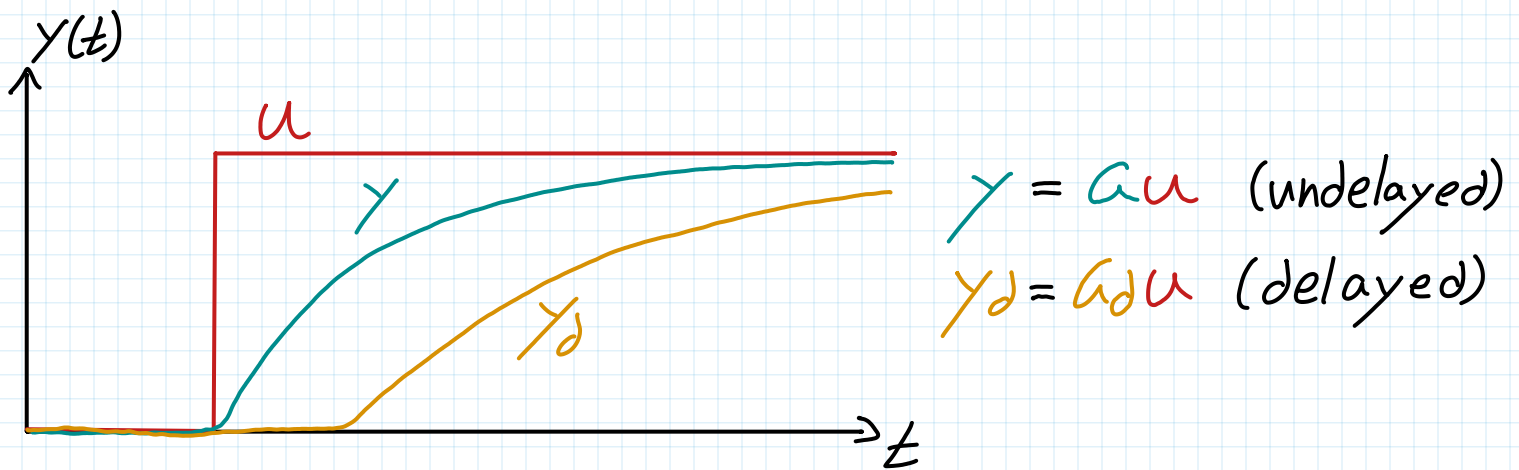


• For an entire length T , we try to control the system with an "old" value.



Time Delays

- To make sure time delays are accounted for, we must analyze their effects in the modeling stage.
- ⇒ Let's begin by deriving how we could mathematically describe these delays



- The underlying system dynamics remain, but the response of A_d is delayed

How do we make A 's and A_d 's only difference the delay?

⇒ Trick: Model as if input $u(t)$ arrives later!

$$y(t) = Au(t) \quad (\text{undelayed})$$

$$y_d(t) = Au(t-T) \quad (\text{delayed})$$

⇒ delayed input by T !

• In the simplest case $u(t) = e^{st}$, so

$$y_d = G e^{s(t-T)} = G e^{st-ST} = G e^{-sT} e^{-st}$$

$$\Rightarrow y_d = \underbrace{G e^{-sT}}_{\text{delayed system } G_d} e^{st} = G_d e^{st} = G_d u(t)$$

Conclusion: e^{-sT} is a delay element

For example, we could describe a delayed open-loop as:

$$L(s) = \frac{1}{s+1} \underbrace{e^{-sT}}$$

This has no poles/zeros!

$\Rightarrow$ Delayed system is not rational

Def: A tf is rational if it can be written as a fraction of two polynomials.

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_0} + d$$

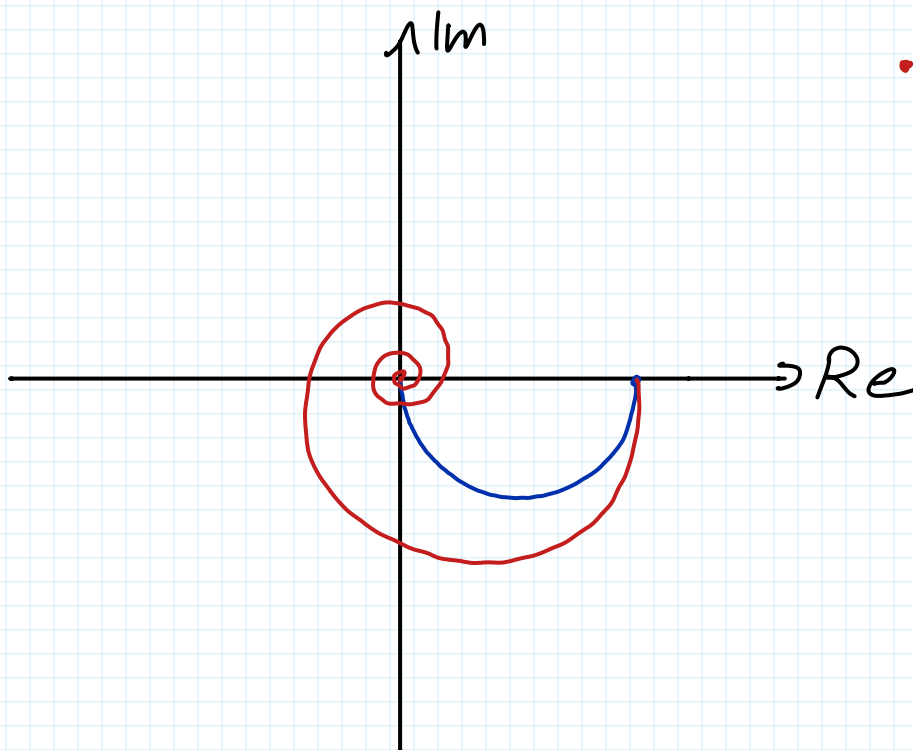
- The lack of rationality poses a problem:

⇒ The root-locus can't be used for stability analysis anymore.

- But we know, that the Nyquist criterion always works.

So what does the Nyquist plot of a delayed system look like?

- Time delays curl the Nyquist-plot of a plant around the origin



- undelayed
- delayed

- Recall: For stability, we count encirclements of the -1 point (or $-1/k$). So the delay intensity will be crucial!

Before learning how this is dealt with, let's take a look at how Bode plots are affected by delays.

$$e^{-sT} = e^{-j\omega T} = \cos(-\omega T) + j\sin(-\omega T)$$

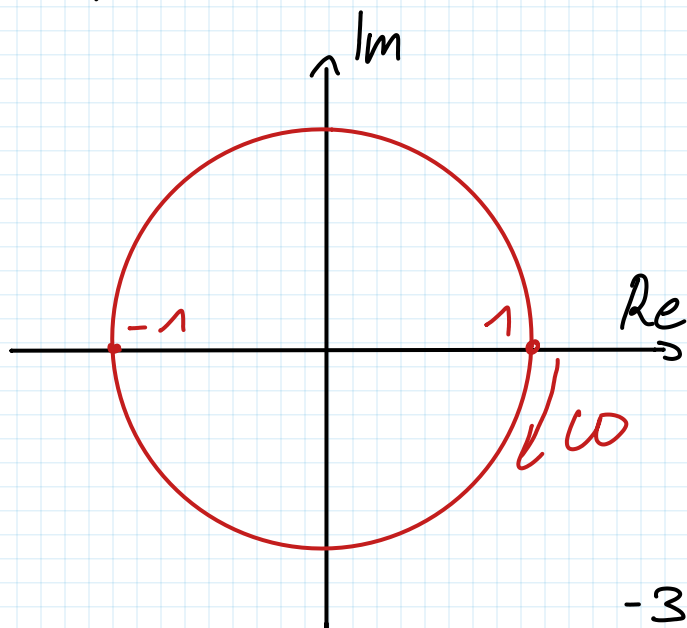
↑
eulers formula

• From this we can extract magnitude/phase

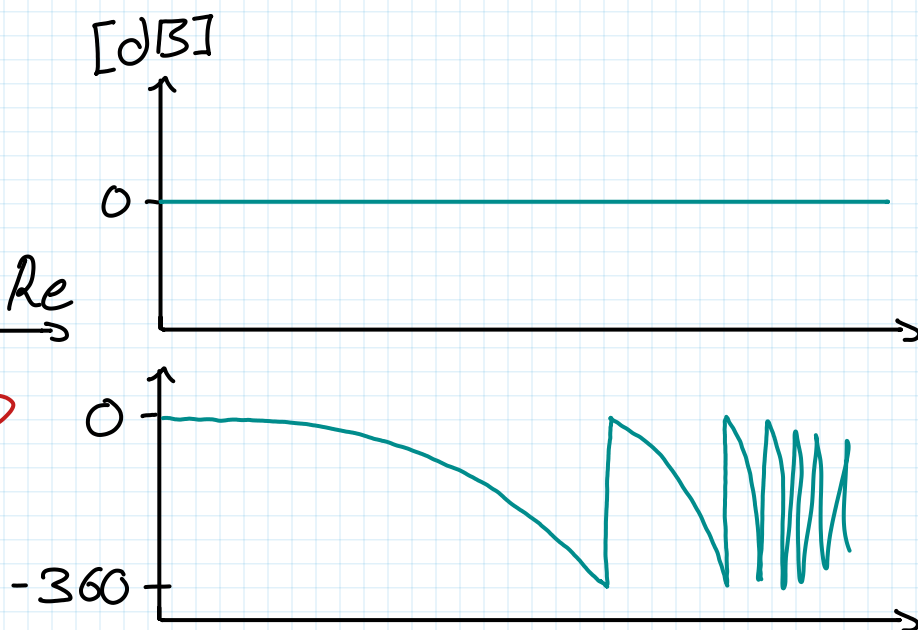
$$\cdot |e^{-sT}| = \sqrt{\cos^2(-\omega T) + \sin^2(-\omega T)} = 1, \quad (0 \text{ dB})$$

$$\begin{aligned} \cdot \angle e^{-sT} &= \angle (\cos(-\omega T) + j\sin(-\omega T)) = \tan^{-1} \left(\frac{-\sin(\omega T)}{\cos(\omega T)} \right) \\ &= \tan^{-1}(-\tan(\omega T)) = -\omega T \end{aligned}$$

• Nyquist:



• Bode

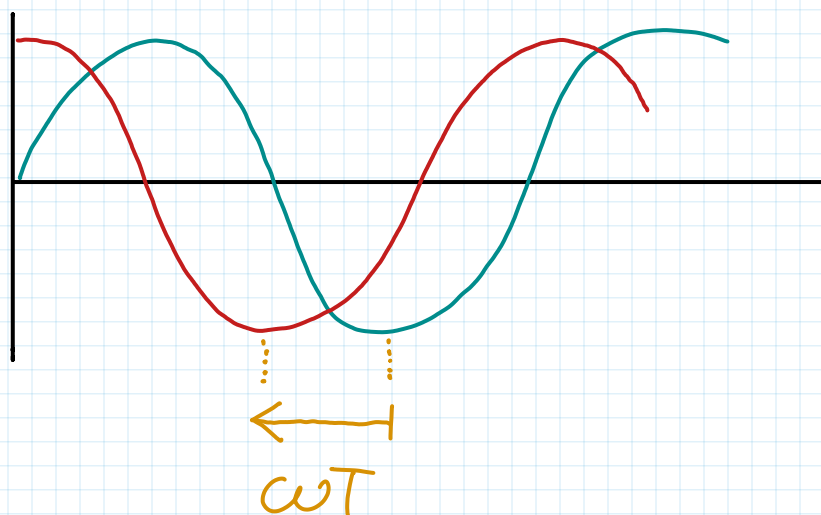


delay has no effect

$$\bullet |L_d(s)| = |L(s)|$$

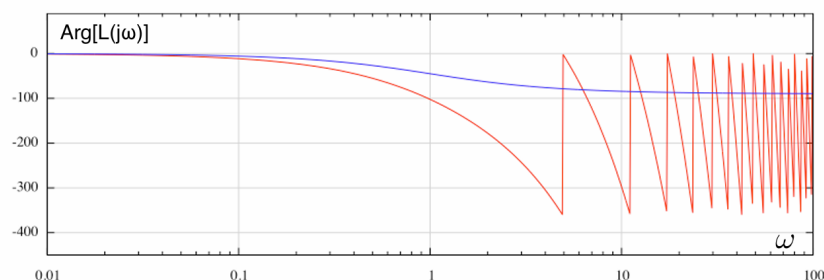
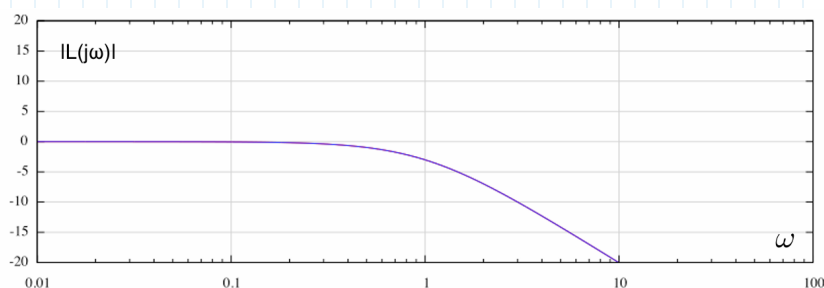
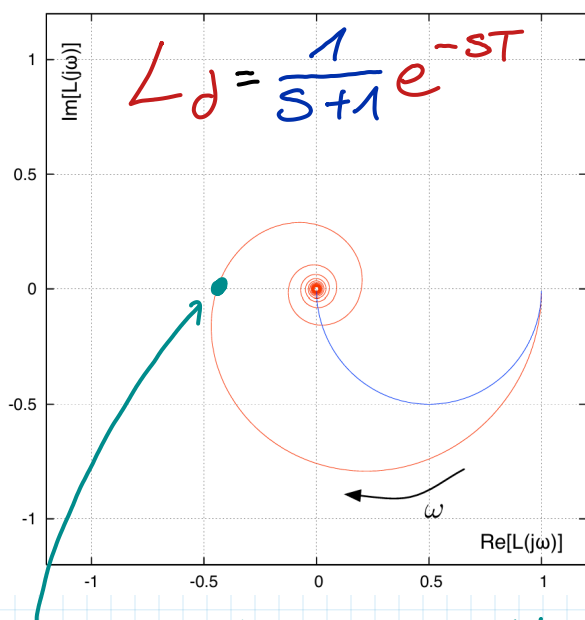
delay shifts by ωT

$$\bullet \angle L_d(s) = \angle L(s) - \omega T$$



same amplitude

Applied to a plant we see this change:



location of this point is relevant for closed-loop stability.

- If we know the delay T , we can find this first crossing by setting:

$$\begin{aligned}\angle L_d(s) &= \angle L(s) - \omega T = \angle \left(\frac{k}{j\omega + 1} \right) - \omega T \\ &= \angle k - \angle(j\omega + 1) - \omega T = -\tan^{-1}(\omega) - \omega T \\ &\stackrel{!}{=} -180^\circ\end{aligned}$$

$$\Rightarrow \tan(\omega) + \omega T = 180^\circ$$

The location of this point determines the gain-margin.

- Similarly, on the bode plot we can read the phase margin with delay as:

$$\phi_m' = \phi_m - \omega_c T$$

In essence, delays decrease margins and therefore robustness. How can we account for this when designing a controller?

• In the lecture, the following procedure is presented:

1) Design a controller for $P(s)$

2) Check phase margin of $P_d(s)$ with:

$$\phi_m' = \phi_m - \omega_c T$$

3) If ϕ_m' too small, add a lead-element

4) Repeat 2 & 3.

This is a very "duct-tape-engineering" - approach... Is there another way to solve this? Yes!

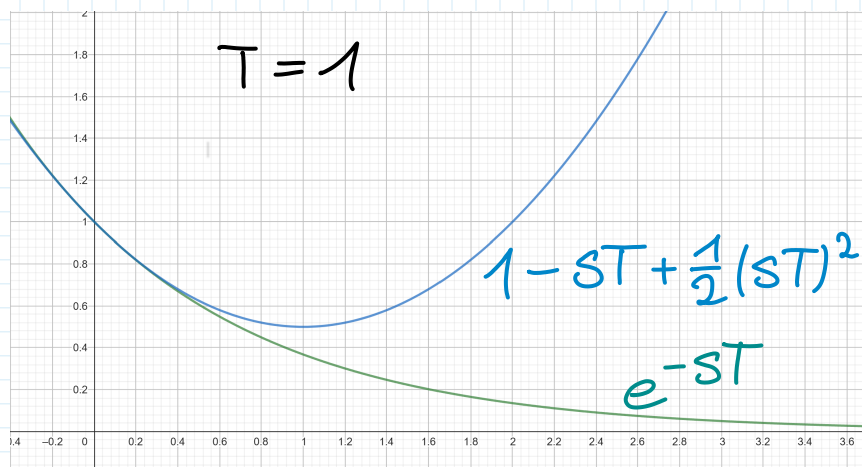
• We can use some mathematical tricks to change the representation of a delay.

⇒ Technically we can write e^{-sT} as a polynomial! Taylor-approximation:

$$e^{-sT} = 1 - sT + \frac{1}{2}(sT)^2 - \frac{1}{6}(sT)^3 \dots$$

An infinite series is hard to implement, therefore we approximate:

$$e^{-sT} \approx 1 - sT + \frac{1}{2}(sT)^2$$



solid for small ω , but bad approximation for $\omega \rightarrow \infty$

A better result is achieved by approximating the delay by a ratio of polynomials. This is the Padé-approximation.

$$e^{-sT} \approx k \cdot \frac{s+p}{s+q} \quad \text{This we can rewrite:}$$

$$k \cdot \frac{s+p}{s+q} = 1 - sT + \frac{1}{2}(sT)^2 - \frac{1}{6}(sT)^3 \dots$$

$$\Rightarrow ks + kp = s + q - s^2T - q sT + \frac{1}{2}s^3T^2 + \frac{1}{2}q(sT)^2 \dots$$

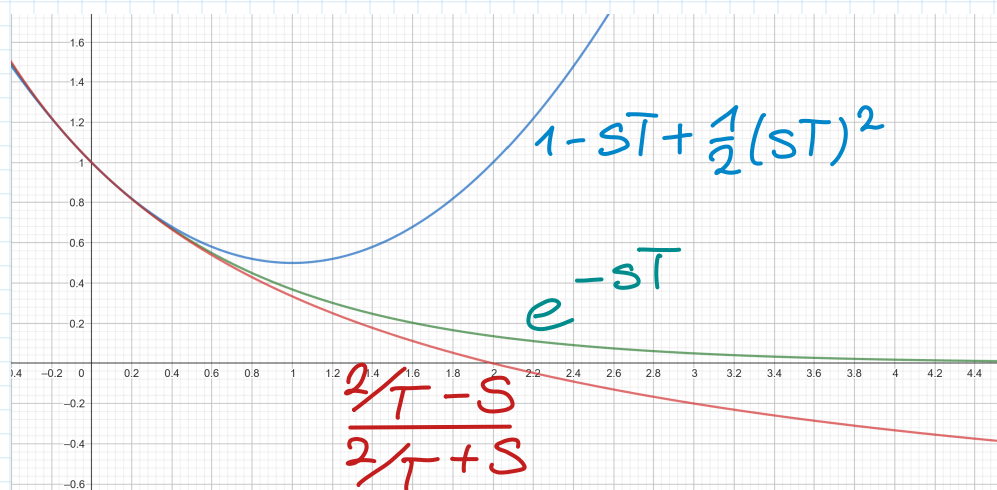
Now we match the coefficients:

- $kp = q$ (0th order)
- $k = 1 - qT$ (1st order)
- $0 = -T + \frac{1}{2}qT^2$ (2nd order)

$\Rightarrow$ Solving this system of equations we get:

$$q = \frac{2}{T}, \quad k = -1, \quad p = -\frac{2}{T}$$

Hence: $e^{-sT} \approx k \cdot \frac{s+p}{s+q} \approx \frac{\frac{2}{T} - s}{\frac{2}{T} + s}$



Think about what this does to our problem.

↪ poles/zeros!!!

$$e^{-sT} \approx \frac{\frac{2}{T} - s}{\frac{2}{T} + s}$$

↪ No poles/zeros

Now we can use a root-locus.

- Using Padé approximations and then checking the root-locus is more of a feasibility check.

⇒ In practice, controllers are mostly designed in discrete time...

- CSII, Signals & Systems

- Step-by-step our controller design becomes more realistic.

⇒ Next week we will look at how we deal with saturations and other "non-neglectable" nonlinearities...

Before we leave the field of linear control theory, let's demonstrate its power.

Old exam question (FS24)

Q37 (1 Points) Mark all correct statements.

- ☒ Suppose that an open loop system is stable. A standard feedback loop is added to the open-loop system such that it is now a closed-loop system. It is not guaranteed that the closed-loop system is stable.
- ☐ Consider a stable closed-loop system. During real-world tests, it was observed that the closed-loop system is subject to a time-delay. If the closed-loop system, without the time delay, is stable, the time-delay cannot affect the stability of the time-delayed closed-loop system
- ☐ Since the transfer function of a pure time delay is a rational function, the stability of a time delayed system can be analyzed using the root locus method.
- ☒ The Padé approximation is a method to approximate the transfer function of a pure time delay, and it can be used in the root locus method to analyze the stability of the approximated system.

(HS23)

Q39 (1 Points) Mark all correct statements.

- ☒ Time delays are linear systems/operators.
- ☒ Let $L(s)$ represent the transfer function of a linear time-invariant system without delays. Let $\tilde{L}(s)$ represent the transfer function of the same system as $L(s)$, where the output of the system now is delayed by 5s, i.e. $\tilde{L}(s)$ is a delayed version of the system represented by $L(s)$. It holds that, $\forall \omega \in \mathbb{R}_{\geq 0}$,

$$\begin{aligned} |\tilde{L}(j\omega)| &= |L(j\omega)|, \text{ and,} \\ \angle \tilde{L}(j\omega) &= \angle L(j\omega) - 5\omega. \end{aligned}$$

- ☐ The transfer function of a time delay is a rational transfer function.
- ☒ Time delays can cause closed-loop systems to become unstable.
- ☒ Time delays can cause significant performance limitations to the achievable bandwidth of closed-loop systems, i.e. the maximal bandwidth while retaining closed-loop stability. More specifically, when a time delay is introduced to a closed-loop system, the achievable bandwidth decreases.

Finally, some quiz-questions!

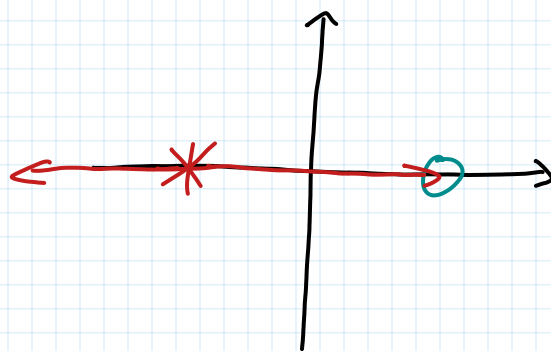
Q1) Given a system $G(s) = \frac{1}{s+1}$.

a time delay $T=2$ affects the system. When using a Padé approx. and a controller k , are there choices of k that make the closed-loop unstable?

☒ TRUE ☐ FALSE

$$P'(s) = k \cdot \frac{1}{s+1} \cdot \frac{1-s}{1+s}$$

$$= k \cdot \frac{1-s}{(1+s)^2}$$



Q2) The magnitude plot is decreased by $-20^{\circ}\text{B}/\text{dec}$ when a delay is introduced.

☐ TRUE ☒ FALSE

Q3) A system $P(s)$ has a crossover frequency $\omega_c = 10 \text{ rad/s}$ and phase margin $\phi_m = 0,2 \text{ rad}$. When adding a delay $T = 0.05$, the phase margin of $P'(s)$ becomes:

$$\phi'_m = \phi_m - \omega_c \cdot T = 0,2 - 10 \cdot 0.05 = -0.3$$

$$\phi'_m = -0,3 \text{ rad}$$